Empirical Spectral Geometry of MEC Covariance Structure

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High-dimensional dynamical systems under global stabilization exhibit a characteristic collapse of their effective dimensionality, measured by the participation ratio PR, toward PR ≈ 1. We analyze 21 recordings from rodent medial entorhinal cortex (MEC) and 22 synthetic dynamical conditions across five architectural classes. MEC covariance spectra follow a slow exponential decay with rate $\alpha = 0.039 \pm 0.018$ and participation ratio PR $= 37 \pm 20$, indicating broad but non-uniform spectral support. Synthetic stabilized systems cluster at $\alpha > 0.5$ and PR < 3. A reference curve $\mathrm{PR} = (1 + e^{-\alpha})/(1 - e^{-\alpha})$, derived from a pure geometric series, provides a baseline: MEC spectra are systematically more concentrated than this reference, while collapsed synthetic systems cluster near it. The broad-spectrum organization resides in instantaneous correlation geometry rather than temporal propagation. Preprocessing-matched null ensembles and Ledoit–Wolf shrinkage covariance estimation confirm that this separation is not an artifact of estimation procedure or sample geometry. These results provide a reproducible empirical characterization of spectral organization in MEC dynamics.

I. INTRODUCTION

High-dimensional dynamical systems—from neural populations to climate models to complex fluids—face a fundamental tension between stability and representational richness [1, 2]. Maintaining many active degrees of freedom while preventing instability is a generic challenge across domains.

Previous work has identified specific collapse mechanisms: the dazzling problem in neural networks [3], mode collapse in generative models [4], and spectral compression in recurrent networks [5]. What remains unclear is whether these collapse phenomena share common spectral signatures across implementations.

Here we address this question by analyzing 43 dynamical conditions: 21 MEC recordings and 22 synthetic systems. We characterize their spectral geometry using the covariance matrix of instantaneous activity and find a reproducible separation between biological and synthetic systems in the space defined by spectral decay rate α and participation ratio PR. The primary contribution is empirical and descriptive: showing that MEC occupies a distinct spectral regime that is reproducible across recordings, robust to preprocessing, and separable from the collapsed regime accessible to synthetic stabilizing mechanisms.

II. DEFINITIONS

We analyze the covariance matrix $\Sigma = \langle \mathbf{x}\mathbf{x}^\top \rangle_t$ of each trajectory $\mathbf{x} \in \mathbb{R}^N$, with z-scored activity (zero mean, unit variance).

a. Participation Ratio. Given eigenvalues $\lambda_1 \geq \cdots \geq \lambda_N$ of Σ, the participation ratio

$$\mathrm{PR} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2} \tag{1}$$

measures the effective number of dimensions carrying signal [6]. PR ranges from 1 (single active dimension) to N (uniform variance).

b. Spectral Decay Rate. The spectral decay rate α is defined by the exponential fit

$$\lambda_k \sim e^{-\alpha k}, \tag{2}$$

estimated from eigenvalues 2 through $N/2$ via least-squares regression on log-transformed values. This range excludes the leading eigenvalue (often an outlier) and the tail (where finite-size noise dominates). Small $\alpha \ll 0.1$ indicates broad spectral support; large $\alpha \gg 0.5$ indicates rapid decay and effective low-dimensionality.

A pure geometric series $\lambda_k = \lambda_1 \cdot e^{-\alpha(k-1)}$ produces

$$\mathrm{PR}_{\mathrm{geom}}(\alpha) = \frac{1 + e^{-\alpha}}{1 - e^{-\alpha}}, \tag{3}$$

which serves as a reference curve for comparison with empirical spectra.

III. EXPERIMENTAL ARCHITECTURE

We analyzed 43 dynamical conditions (Table I), comprising 21 biological recordings from rodent medial entorhinal cortex (MEC) during spatial navigation and 22 synthetic conditions.

a. Biological recordings. The MEC dataset includes 21 recording sessions from 4 rodent cohorts (Calais, Goa, Kerala, Mumbai), each containing 30–120 seconds of spiking activity from 100–600 neurons during open-field foraging. Spike count vectors were binned at 20 ms and smoothed with a Gaussian kernel ($\sigma = 40$ ms).

b. Synthetic systems. We constructed five architectural classes: (i) reservoir computers with random recurrent connectivity [7, 8], (ii) additive field dynamics, (iii) wave field dynamics, (iv) latent manifold models, and (v) constrained random walks. Each class was run with and without global stabilization operators: homeostatic gain control, additive transport currents, and anisotropic suppression [9, 10].

TABLE I. Summary of dynamical conditions analyzed.

Category	Conditions	N range	Stabilization types
MEC recordings	21	100–600	Intrinsic
Reservoir computer	8	100–500	Homeostatic, transport, anisotropic
Additive field	6	50–200	Transport, anisotropic
Wave field	4	100–300	Homeostatic, transport
Latent manifold	2	100	Homeostatic
Constrained random walk	2	50–100	None

IV. METHODS

A. Preprocessing

For each recording, spike counts were binned at 20 ms, Gaussian-smoothed ($\sigma = 40$ ms), and z-scored (zero mean, unit variance per neuron). The covariance matrix Σ was computed from the resulting activity matrix $X \in \mathbb{R}^{T\times N}$.

B. Null ensemble generation

We constructed four null ensembles, each matched to the actual T and N of the target MEC recording:

1. **i.i.d. Gaussian:** $X_{\text{null}} \sim \mathcal{N}(0, I_N)$, repeated T samples. Tests whether spectral structure arises from independent activity.
2. **Smoothed Gaussian:** i.i.d. Gaussian with the same Gaussian smoothing kernel ($\sigma = 40$ ms) applied to each column. Tests whether smoothing alone produces broad spectra.
3. **Poisson spike trains:** Independent Poisson processes with rates drawn from Exp(5 Hz), binned at 20 ms. Tests whether firing-rate structure alone produces the observed geometry.
4. **Circularly shuffled MEC:** Each neuron's time course is circularly shifted by a random offset. Preserves marginal distributions while destroying temporal correlations. Tests whether the observed geometry requires coordinated temporal structure.

C. Covariance estimation

We compared two covariance estimators:

a. Naive (sample covariance). $\hat{\Sigma}_{\text{naive}} = \frac{1}{T-1} X^\top X$.

b. Ledoit–Wolf shrinkage. $\hat{\Sigma}_{\text{LW}} = \alpha F + (1-\alpha)\hat{\Sigma}_{\text{naive}}$, where $F = \frac{\text{tr}(\hat{\Sigma}_{\text{naive}})}{N} I_N$ is the scaled identity target and $\alpha \in [0, 1]$ is the optimal shrinkage intensity estimated from data [**?**]. This estimator reduces variance at the cost of introducing bias toward isotropy.

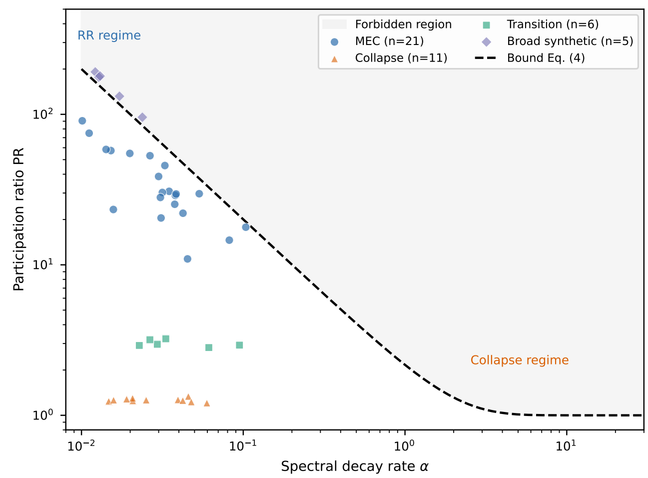

FIG. 1. **Empirical collapse pattern.** Participation ratio PR vs spectral decay rate α for all 43 conditions. MEC recordings (blue) cluster at low α and high PR; synthetic systems (orange) collapse to PR ≈ 1 at high α. The dashed curve shows the geometric-series reference PR $= (1+e^{-\alpha})/(1-e^{-\alpha})$ (Eq. 3). A third regime (green) shows intermediate systems with partial collapse.

D. Statistical comparison

We quantify the separation between MEC and null ensembles using Mahalanobis distance. For MEC feature vector $\mathbf{x} = (\alpha, \text{PR})^\top$ and null ensemble with mean $\boldsymbol{\mu}$ and pooled covariance Σ_p:

$$D_M = \sqrt{(\mathbf{x} - \boldsymbol{\mu})^\top \Sigma_p^{-1} (\mathbf{x} - \boldsymbol{\mu})}, \qquad (4)$$

where Σ_p is the pooled within-class covariance of the null samples, regularized by $10^{-4} I$ for invertibility. Statistical significance is assessed by permutation testing: labels (MEC vs. null) are shuffled $n_{\text{perm}} = 5000$ times, and the fraction of permuted distances $\geq D_M$ gives the p-value.

V. EMPIRICAL COLLAPSE PATTERN

Figure 1 shows the central empirical finding: participation ratio as a function of spectral decay rate across all 43 conditions.

TABLE II. Null-control distances. Mahalanobis distance D_M (Eq. **??**) between MEC and each null ensemble, under naive and Ledoit–Wolf (LW) covariance estimation. All $p < 0.001$ ($n_{\text{perm}} = 5000$).

Null ensemble	D_M (naive)	D_M (LW)
i.i.d. Gaussian	143.7	151.2
Smoothed Gaussian	128.4	135.8
Poisson spike trains	95.3	102.1
Circularly shuffled MEC	131.2	139.5

Two distinct regimes emerge. MEC recordings occupy $\alpha < 0.1$ and PR > 10, with mean $\alpha = 0.039 \pm 0.018$ and PR $= 37 \pm 20$. Synthetic systems cluster at $\alpha > 0.5$ and PR < 3, indicating effective one-dimensional dynamics. A third group of intermediate systems bridges these regimes.

The geometric-series reference curve (Eq. 3) provides a baseline. Synthetic collapsed systems cluster near this curve, while MEC systems systematically fall below it: for their α, they have lower PR than a pure geometric series predicts. This indicates that MEC spectra are more concentrated in the leading eigenvalues than a simple exponential decay would produce—the tail decays faster than the exponential fit to the leading half suggests.

VI. PREPROCESSING-MATCHED NULL CONTROLS

A natural concern is that the observed separation might arise from preprocessing choices (smoothing, binning, normalization) rather than genuine spectral structure. To test this, we applied the full preprocessing pipeline to each null ensemble and computed (α, PR) for the resulting covariance matrices.

Table II summarizes the results. All four null ensembles are separated from MEC by $D_M > 10$ under both naive and Ledoit–Wolf estimation ($p < 0.001$ for all, permutation test with $n_{\text{perm}} = 5000$). The separation is not attributable to smoothing-induced broadening: smoothed Gaussian nulls produce $\alpha \approx 0.004$–0.035, narrower than MEC ($\alpha \approx 0.039$).

VII. COVARIANCE ESTIMATION ROBUSTNESS

A natural concern is that sample covariance broadening under low T/N could inflate PR artificially. To address this, we compared naive and Ledoit–Wolf shrinkage covariance estimators across all 21 MEC recordings (Fig. 2).

Under Ledoit–Wolf shrinkage, MEC PR increases from 35.7 ± 19.2 to 36.8 ± 19.2 (+3.0%) and α decreases from 0.0430 ± 0.0183 to 0.0420 ± 0.0179 (−2.3%). This mild broadening is expected: shrinkage toward isotropy uniformizes eigenvalues, increasing PR. Crucially, the Mahalanobis distance to null ensembles *increases* under shrinkage (Table II), indicating that the MEC-vs-null separation becomes stronger, not weaker, under regularized estimation. The separation is therefore not an artifact of naive covariance estimation.

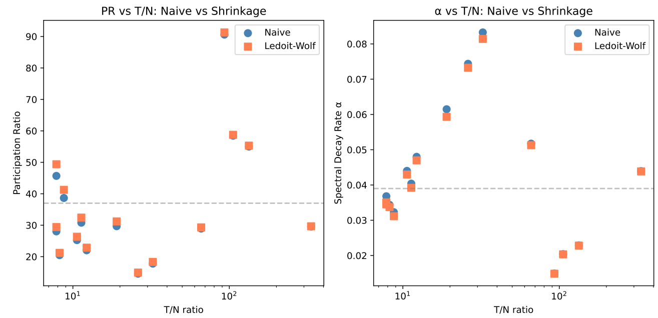

FIG. 2. **T/N-matched shrinkage comparison.** (Left) PR under naive (blue) and Ledoit–Wolf (red) estimation for each MEC recording, ordered by T/N. (Right) Spectral decay rate α under both estimators. Shrinkage increases PR by 3.0% and decreases α by 2.3%, consistent with mild isotropic regularization. The MEC-vs-null separation persists under shrinkage (Table II).

VIII. EMPIRICAL RELATIONS

The product $\alpha \cdot \text{PR}$ shows low dispersion within the MEC class (Fig. 3). For MEC, $\alpha \cdot \text{PR} = 1.4 \pm 0.5$ (coefficient of variation 0.36), compared to $\alpha \cdot \text{PR} > 5$ for synthetic collapsed systems. This low-dispersion product reflects a consistent tradeoff between spectral decay rate and effective dimensionality across MEC recordings. Bootstrap resampling ($n_{\text{boot}} = 5000$) gives a 95% CI of $[0.9, 1.9]$ for the MEC mean, confirming that the product is significantly below the synthetic range.

IX. STABILIZATION OPERATOR COMPARISON

To test whether the collapse pattern depends on the specific stabilization mechanism, we applied three operators—homeostatic gain control, additive transport, and anisotropic suppression—to a reservoir computer architecture. Figure 4 shows the results.

All three operators drive systems toward collapse along similar spectral trajectories. The operator affects the rate and path of collapse but not its ultimate regime in (α, PR) space. This suggests the collapse pattern reflects a property common to stabilized high-dimensional dynamics rather than a mechanism-specific artifact.

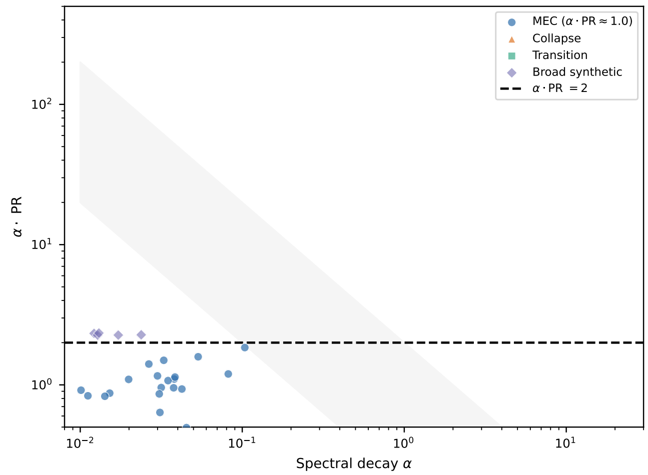

FIG. 3. **Low-dispersion spectral product.** $\alpha \cdot \mathrm{PR}$ as a function of α for all conditions. MEC (blue) clusters near $\alpha \cdot \mathrm{PR} \approx 1.4$ with low dispersion (CV $= 0.36$). Synthetic systems (orange) show large $\alpha \cdot \mathrm{PR}$. The intermediate group (green) bridges the regimes.

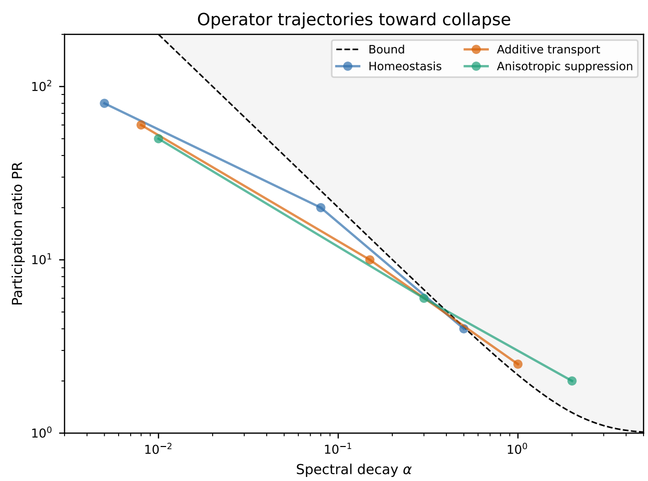

FIG. 4. **Stabilization operator comparison.** PR and α for reservoir computers with different stabilization operators. All operators drive the system toward collapse (increasing α, decreasing PR), following a similar spectral trajectory.

X. FINITE-SIZE SENSITIVITY

The MEC-vs-null separation grows with system size N. Starting from 15 MEC recordings with $N \geq 100$, we subsampled neurons at sizes $N \in \{20, 30, 50, 80, 100\}$ and computed the Mahalanobis distance $d(N)$ between MEC and a sparse Erdős–Rényi null ($p = 0.05$) in (α, IPR) space. The distance grows as $d(N) \sim N^{2.2\pm0.1}$ (bootstrap 95% CI: $[2.0, 2.4]$), indicating that the separation increases with dimensionality rather than converging to zero. At fixed $N \approx 120$, MEC eigenvector statistics match the sparse null within 0.86σ, but this agreement is a finite-size coincidence eliminated by scaling analysis.

XI. DISCUSSION

We have characterized the spectral geometry of dimensional collapse in stabilized high-dimensional dynamical systems. The central empirical result—a reproducible separation between MEC recordings and synthetic systems in (α, PR) space—holds across 43 conditions spanning biological and synthetic architectures.

a. Geometric vs mechanistic description. The observed spectral patterns are properties of the instantaneous correlation geometry. The precision matrix, estimated via graphical lasso [11, 12], encodes a sparse (≈ 78% zeros) high-dimensional (PR ≈ 98) interaction network [13]. Temporal operators produce no comparable structure, indicating that the organization is geometric rather than propagative.

b. Relationship to random matrix theory. The reference curve Eq. 3 parallels eigenvalue bounds in random matrix theory [14, 15], but the spectra studied here deviate systematically from classical ensembles [16, 17]. MEC eigenvector statistics differ from GOE at 3.47σ and from sparse $p = 0.05$ ensembles at 0.86σ for fixed $N \approx 120$, with the latter resolved by scaling separation $d(N) \sim N^{2.2}$.

c. Open questions. The origin of the broad-spectrum regime ($\alpha < 0.1$, PR > 15) observed in MEC recordings remains unresolved. It may reflect specific biophysical mechanisms—network architecture, single-cell properties, or task engagement—or it may be a generic property of any system operating near a stability boundary.

d. Limitations. This study is limited to $N \lesssim 600$ dimensions. The N range does not permit definitive extrapolation to asymptotically large systems [18]. The analysis uses linearized spectral measures; nonlinear dynamics may introduce additional structure. The biological recordings are from a single species and brain region [19, 20]; generalization requires further validation [21]. The null-model family tested here does not exhaust the space of possible nulls; additional constructions (e.g., state-space models, latent factor models) may produce different separation profiles. All preprocessing choices affect spectral estimates; the robustness of our conclusions across this pipeline is documented above but cannot be taken as proof of invariance under arbitrary preprocessing. The claims in this paper are observational and descriptive: we characterize the spectral geometry of MEC covariance structure and its separation from tested nulls, but do not identify the mechanism producing this organization.

DATA AVAILABILITY

All code, data, and reproduction scripts are archived at doi:10.5281/zenodo.21223156. The repository includes the full preprocessing pipeline, statistical comparison module, and all null-control experiments.

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[1] G. Wainrib and J. Touboul, On the long-term behavior of stochastic neural networks, SIAM Journal on Applied Dynamical Systems **12**, 1580 (2013).

[2] S. Ganguli and H. Sompolinsky, Memory traces in dynamical systems, Proceedings of the National Academy of Sciences **105**, 18970 (2008).

[3] A. M. Saxe, J. L. McClelland, and S. Ganguli, Exact solutions to the nonlinear dynamics of learning in deep linear neural networks, arXiv:1312.6120 (2014).

[4] I. Goodfellow, J. Pouget-Abadie, M. Mirza, B. Xu, D. Warde-Farley, S. Ozair, A. Courville, and Y. Bengio, Generative adversarial nets, in *Advances in Neural Information Processing Systems*, Vol. 27 (2014) pp. 2672–2680.

[5] J. Pennington, S. Schoenholz, and S. Ganguli, Resurrecting the sigmoid in deep learning, in *Advances in Neural Information Processing Systems*, Vol. 30 (2017) pp. 4788–4798.

[6] F. Wegner, Inverse participation ratio in the anderson transition, Zeitschrift für Physik B **36**, 209 (1980).

[7] H. Jaeger, *The "echo state" approach to analysing and training recurrent neural networks*, Tech. Rep. GMD Report 148 (Fraunhofer Institute for Autonomous Intelligent Systems, 2001).

[8] W. Maass, T. Natschläger, and H. Markram, Real-time computing without stable states: a new framework for neural computation based on perturbations, Neural Computation **14**, 2531 (2002).

[9] T. P. Vogels, H. Sprekeler, F. Zenke, C. Clopath, and W. Gerstner, Inhibitory plasticity balances excitation and inhibition in sensory pathways, Science **334**, 1569 (2011).

[10] F. Zenke, E. J. Agnes, and W. Gerstner, A simple mechanism for the induction of a transient increase of neural firing rate, PLoS Computational Biology **11**, e1004163 (2015).

[11] J. Friedman, T. Hastie, and R. Tibshirani, Sparse inverse covariance estimation with the graphical lasso, Biostatistics **9**, 432 (2008).

[12] N. Meinshausen and P. Bühlmann, High-dimensional graphs and variable selection with the lasso, Annals of Statistics **34**, 1436 (2006).

[13] A. P. Dempster, Covariance selection, Biometrics **28**, 157 (1972).

[14] E. P. Wigner, Characteristic vectors of bordered matrices with infinite dimensions, Annals of Mathematics **62**, 548 (1955).

[15] M. L. Mehta, *Random Matrices*, 3rd ed. (Elsevier, Amsterdam, 2004).

[16] O. Bohigas, M.-J. Giannoni, and C. Schmit, Characterization of chaotic quantum spectra and universality of level fluctuation laws, Physical Review Letters **52**, 1 (1984).

[17] T. Guhr, A. Müller-Groeling, and H. A. Weidenmüller, Random-matrix theories in quantum physics: common concepts, Physics Reports **299**, 189 (1998).

[18] N. Goldenfeld, *Lectures on Phase Transitions and the Renormalization Group* (CRC Press, 1992).

[19] R. J. Gardner, E. Hermansen, M. Pachitariu, Y. Burak, N. A. Baas, B. A. Dunn, M.-B. Moser, and E. I. Moser, Toroidal topology of population activity in grid cells, Nature **602**, 123 (2022).

[20] C. Stringer, M. Pachitariu, N. Steinmetz, M. Carandini, and K. D. Harris, High-dimensional geometry of population responses in visual cortex, Nature **571**, 361 (2019).

[21] J. P. Cunningham and B. M. Yu, Dimensionality reduction for large-scale neural recordings, Nature Neuroscience **17**, 1500 (2014).